

CCNA COMMAND REFERENCE

Cisco Certified Network Associate (200-120)

Complete extracted IOS command guide
Compiled for print & offline study

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ROUTER BASIC COMMANDS

Command	Description
Router>enable	To enter to the Privileged mode
Router>ping <ip address>	To ping IP or device
Router>tracertoute <ip>	To trace route to any remote device
Router>telnet <ip>	To telnet any remote device
Router>sh clock	To view date and time
Router>sh users	To view the remote users
Router>sh ip int brief	To display status of all interfaces
Router>sh flash	To display the status of Flash memory
Router>sh history	To display the command history
Router>sh version	To display the system hardware and software status
Router>ssh	To access remote device through SSH
Router>logout	To logout from router
Router>exit	To go back one step
Router#copy run start	To save the configuration to NVRAM
Router#wr	To save the configuration
Router#clock set hh:mm:ss 24 aug 2013	To set date and time
Router#reload	To restart router
Router#config terminal	To enter to configuration mode
Router#sh running-config	To display the entire router configuration
Router#sh hosts	To display host table
Router#sh arp	To display arp table
Router#sh ip route	To display routing table
Router#sh protocols	To list all protocols
Router#sh ip protocols	To list all configured routing protocols
Router#sh interfaces	To list all interfaces
Router#sh access-lists	To display access-lists
Router#sh clock	To list clock status

ROUTER BASIC CONFIGURATION

Command	Description
Router(config)#hostname <new router name>	To change router name
Router(config)#banner motd	To set banner
Router(config)#service password-encryption	To encrypt passwords
Router(config)#interface <interface name> Router(config-if)#ip add <ip address> <subnet mask>	To assign IP address on interface
Router(config)#interface <interface name> Router(config-if)#no ip add	To remove IP address from interface
Router(config)#interface <interface name> Router(config-if)#no shutdown	To enable interface
Router(config)#interface <interface name> Router(config-if)#shutdown	To disable interface
Router(config)#interface <interface name> Router(config-if)#clock rate 64000	To set clock rate on serial interface
Router(config)#ip host <hostname> <ip address>	To create host table
Router(config)#username <username> password <password>	To create user and set password on it

SETTING UP ROUTER PASSWORDS

Command	Description
Router(config)#enable password <password>	To set enable password
Router(config)#enable secret <password>	To set enable secret
Router(config)#line con 0 Router(config-line)#password <password> Router(config-line)#login	To set line console password
Router(config)#line con 0 Router(config-line)#login local	To set users password on console
Router(config)#line vty 0 4 Router(config-line)#password <password> Router(config-line)#login	To set line VTY password

REMOVING ROUTER PASSWORDS

Command	Description
Router(config)#no password <password>	To remove enable password
Router(config)#no enable secret	To remove enable secret
Router(config)#line con 0 Router(config-line)#no password Router(config-line)#no login	To remove line console password
Router(config)#line con 0 Router(config-line)#no login local	To remove users password from line console
Router(config)#line vty 0 4 Router(config-line)#no password Router(config-line)#no login	To remove line VTY password

ROUTER BACKUP & RESTORE

Types of backup: NVRAM backup, FLASH backup.

Command	Description
Router#copy startup-config tftp	NVRAM backup using TFTP
Router#copy flash tftp	FLASH backup using TFTP
Router#copy tftp running-config	NVRAM restore from TFTP
Router(config)#ip ftp username cisco Router(config)#ip ftp password 123 Router#copy startup-config ftp	NVRAM backup using FTP
Router#copy ftp running-config	NVRAM restore from FTP

DHCP SERVER CONFIGURATION COMMANDS

Command	Description
Router(config)#ip dhcp pool <pool name> Example: Router(config)#ip dhcp pool Cisco	To set DHCP pool name
Router(dhcp-config)#network <network address> <subnet mask> Example: Router(dhcp-config)#network 192.168.2.0 255.255.255.0	To define network address and subnet mask
Router(dhcp-config)#default-router <gateway address> Example: Router(dhcp-config)#default-router 192.168.2.254	To define default gateway (up to 8 addresses)
Router(dhcp-config)#dns-server <DNS address> Example: Router(dhcp-config)#dns-server 192.168.2.253	To define DNS address (up to 8 addresses)
Router(dhcp-config)#lease <days> <hours> <minutes> or: Router(dhcp-config)#lease infinite	To define lease duration
Router(dhcp-config)#domain-name <domain name> Example: Router(dhcp-config)#domain-name Cisco.com	To define domain name
Router(config)#ip dhcp excluded-address <start ip> <end ip>	To define DHCP exclusion
Router(config)#ip dhcp ping timeout <milliseconds>	To define DHCP ping timeout (100-10000 ms)

Command	Description
Router(config)#ip dhcp pool <pool name> Router(dhcp-config)#host <host ip> Router(dhcp-config)#hardware-address <mac> Router(dhcp-config)#client-name <hostname>	To define DHCP reservation
Router#sh ip dhcp binding	To view addresses assigned to clients
Router#sh ip dhcp binding <client address>	To view a particular client address
Router#clear ip dhcp binding <client address> *	To clear assigned client address
Router#debug ip dhcp server events packets linkage	To troubleshoot DHCP

FULL DHCP CONFIGURATION EXAMPLE

Pool name: Cisco | Range: 192.168.2.1-252 | Network: 192.168.2.0/255.255.255.0 | Gateway: 192.168.2.254 | DNS: 192.168.2.253 | Lease: 10 days | Exclusion: 192.168.2.1-10

Command	Description
Router(config)#ip dhcp pool Cisco Router(dhcp-config)#network 192.168.2.0 255.255.255.0 Router(dhcp-config)#domain-name Cisco.com Router(dhcp-config)#default-router 192.168.2.254 Router(dhcp-config)#dns-server 192.168.2.253 Router(dhcp-config)#lease 10 Router(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.10	Full DHCP pool configuration

STATIC ROUTING

Syntax: Router(config)# ip route <destination network> <subnet mask> <next-hop address>

Command	Description
Router(config)#ip route 192.168.2.0 255.255.255.0 11.0.0.2	Adds a static route to 192.168.2.0/24 via next hop 11.0.0.2

STATIC ROUTING - MULTI-ROUTER EXAMPLE

Four-router internetwork (Router A-D) with LAN and WAN links.

Command	Description
Router A: int fa0/0 -> ip add 192.168.1.254 255.255.255.0, no shutdown int s1/0 -> ip add 11.0.0.1 255.0.0.0, clock rate 64000, no shut ip route 192.168.2.0 255.255.255.0 11.0.0.2 ip route 192.168.3.0 255.255.255.0 11.0.0.2 ip route 192.168.4.0 255.255.255.0 11.0.0.2 ip route 12.0.0.0 255.0.0.0 11.0.0.2 ip route 13.0.0.0 255.0.0.0 11.0.0.2	Router A configuration
Router B: int fa0/0 -> ip add 192.168.2.254 255.255.255.0, no shutdown int s1/0 -> ip add 11.0.0.2 255.0.0.0, clock rate 64000, no shut int s1/1 -> ip add 12.0.0.1 255.0.0.0, clock rate 64000, no shut ip route 192.168.1.0 255.255.255.0 11.0.0.1 ip route 192.168.3.0 255.255.255.0 12.0.0.2 ip route 192.168.4.0 255.255.255.0 12.0.0.2 ip route 13.0.0.0 255.0.0.0 12.0.0.2	Router B configuration
Router C: int fa0/0 -> ip add 192.168.3.254 255.255.255.0, no shutdown int s1/0 -> ip add 12.0.0.2 255.0.0.0, clock rate 64000, no shut int s1/1 -> ip add 13.0.0.1 255.0.0.0, clock rate 64000, no shut ip route 192.168.1.0 255.255.255.0 12.0.0.1 ip route 192.168.2.0 255.255.255.0 12.0.0.1 ip route 192.168.4.0 255.255.255.0 13.0.0.2 ip route 11.0.0.0 255.0.0.0 12.0.0.1	Router C configuration
Router D: int fa0/0 -> ip add 192.168.4.254 255.255.255.0, no shutdown int s1/0 -> ip add 13.0.0.2 255.0.0.0, clock rate 64000, no shut	Router D configuration

Command	Description
ip route 192.168.1.0 255.255.255.0 13.0.0.1 ip route 192.168.2.0 255.255.255.0 13.0.0.1 ip route 192.168.3.0 255.255.255.0 13.0.0.1 ip route 11.0.0.0 255.0.0.0 13.0.0.1 ip route 12.0.0.0 255.0.0.0 13.0.0.1	

RIP CONFIGURATION

Command	Description
Router(config)#router rip Router(config-router)#version <1/2> Router(config-router)#network <network address>	To configure RIP
Router#show ip route	To list routing table

RIP - 7 ROUTER INTERNETWORK EXAMPLE

Routers A-G configured with RIPv2, each advertising its connected networks.

Command	Description
Router A: int fa0/0 -> ip add 192.168.1.254 255.255.255.0, no shutdown int s0/0 -> ip add 11.0.0.1 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 12.0.0.1 255.0.0.0, clock rate 64000, no shut int s0/2 -> ip add 13.0.0.1 255.0.0.0, clock rate 64000, no shut router rip version 2 network 192.168.1.0 network 11.0.0.0 network 12.0.0.0 network 13.0.0.0	Router A
Router B: int s0/0 -> ip add 11.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 14.0.0.1 255.0.0.0, clock rate 64000, no shut router rip version 2 network 11.0.0.0 network 14.0.0.0	Router B
Router C: int s0/0 -> ip add 12.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 15.0.0.1 255.0.0.0, clock rate 64000, no shut router rip version 2 network 12.0.0.0 network 15.0.0.0	Router C
Router D: int s0/0 -> ip add 13.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 16.0.0.1 255.0.0.0, clock rate 64000, no shut router rip version 2 network 13.0.0.0 network 16.0.0.0	Router D
Router E: int fa0/0 -> ip add 192.168.2.254 255.255.255.0, no shut int s0/0 -> ip add 14.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 17.0.0.1 255.0.0.0, clock rate 64000, no shut router rip version 2 network 192.168.2.0 network 14.0.0.0 network 17.0.0.0	Router E

Command	Description
Router F: int s0/0 -> ip add 15.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 17.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/2 -> ip add 18.0.0.1 255.0.0.0, clock rate 64000, no shut router rip version 2 network 15.0.0.0 network 17.0.0.0 network 18.0.0.0	Router F
Router G: int s0/0 -> ip add 16.0.0.2 255.0.0.0, clock rate 64000, no shut int s0/1 -> ip add 18.0.0.2 255.0.0.0, clock rate 64000, no shut router rip version 2 network 16.0.0.0 network 18.0.0.0	Router G

IGRP CONFIGURATION / VERIFICATION

Command	Description
Router#show ip route	Shows the routing table
Router#show ip protocols	Shows routing protocol parameters and current timer values
Router#debug ip igrp transactions	Issues log messages with details of the IGRP updates
Router#debug ip igrp events	Issues log messages for each IGRP update
Router(config)#router igrp <ASN> Router(config-router)#network <network address> Example: Router(config)#router igrp 111 Router(config-router)#network 192.168.1.0	To configure IGRP

EIGRP CONFIGURATION / VERIFICATION

Command	Description
Router#show ip route	Shows the routing table
Router#show ip eigrp topology	Shows feasible successor routes
Router#show ip eigrp topology network	Shows all entries in topology table for given destination network
Router#show ip eigrp topology all-links	Shows all entries in the topology table
Router#show ip eigrp neighbor	Shows the neighbor table
Router(config)#router eigrp <ASN> Router(config-router)#network <network address> Example: Router(config)#router eigrp 111 Router(config-router)#network 192.168.1.0	To configure EIGRP

ADVANCED EIGRP CONFIGURATION

Command	Description
Router(config)#router eigrp <ASN> Router(config-router)#distance eigrp <distance> Example: distance eigrp 122	Configure EIGRP with different Administrative Distance
Router(config)#router eigrp <ASN> Router(config-router)#metric maximum-hops <1-255> Example: metric maximum-hops 255	To extend EIGRP hops
Router(config)#router eigrp <ASN> Router(config-router)#auto-summary	Enable automatic network number summarization
Router(config)#router eigrp <ASN> Router(config-router)#maximum-paths <1-16> Example: maximum-paths 16	Forward packets over multiple paths

Command	Description
Router(config)#router eigrp <ASN> Router(config-router)#variance <1-128> Example: variance 128	Control load balancing variance

OSPF CONFIGURATION / VERIFICATION

Command	Description
Router#show ip route	Shows the routing table
Router#show ip protocols	Shows routing table parameters and current timer values
Router#show ip ospf interface	Lists area and adjacent neighbors
Router#show ip ospf neighbor	Lists neighbors and current status per interface
Router#show ip route ospf	Lists routes learned by OSPF
Router#debug ip ospf events	Issues log messages for each OSPF packet
Router#debug ip ospf packet	Issues log messages describing the contents of all OSPF packets
Router#debug ip ospf hello	Issues log messages describing Hellos and Hello failures
Router(config)#router ospf <ASN> Router(config-router)#network <network address> <wildcard mask> area <area no.> Example: Router(config)#router ospf 111 Router(config-router)#network 192.168.1.0 0.0.0.255 area 0	To configure OSPF

NAT - STATIC NAT

Command	Description
Router(config)#ip nat inside source static 192.168.1.2 12.0.0.3 Router(config)#ip nat inside source static 192.168.1.3 12.0.0.4 Router(config)#interface s1/0 Router(config-if)#ip nat outside Router(config-if)#interface f0/0 Router(config-if)#ip nat inside	Configure static NAT
Router(config)#no ip nat inside source static 192.168.1.2 12.0.0.3 Router(config)#no ip nat inside source static 192.168.1.3 12.0.0.4	Remove static NAT

NAT - DYNAMIC NAT

Command	Description
Router(config)#ip nat pool abc 11.0.0.1 11.0.0.15 netmask 255.0.0.0 Router(config)#access-list 10 192.168.1.0 0.0.0.255 Router(config)#ip nat inside source list 10 pool abc Router(config)#interface s1/0 Router(config-if)#ip nat outside Router(config-if)#interface f0/0 Router(config-if)#ip nat inside	Configure dynamic NAT
Router(config)#no ip nat pool abc Router(config)#no access-list 10 Router(config)#no ip nat inside source list 10	Remove dynamic NAT

NAT - PAT (OVERLOAD)

Command	Description
Router(config)#ip nat pool abc 11.0.0.1 11.0.0.15 netmask 255.0.0.0 Router(config)#access-list 10 192.168.1.0 0.0.0.255 Router(config)#ip nat inside source list 10 pool abc overload Router(config)#interface s1/0 Router(config-if)#ip nat outside	Configure PAT / NAT overload

Command	Description
Router(config-if)#interface f0/0 Router(config-if)#ip nat inside	
Router(config)#no ip nat pool abc Router(config)#no access-list 10 Router(config)#no ip nat inside source list 10	Remove PAT configuration

BASIC SWITCH COMMANDS & CONFIGURATION

Command	Description
Switch>enable	To log in to privileged mode
Switch#conf t	To enter configuration mode
Switch#telnet <ip address>	To access remote device through Telnet
Switch#ping <ip address>	To check connectivity to the remote device
Switch#sh run	To display the entire configuration
Switch#copy run start	To save the configuration into NVRAM
Switch#wr	To save the configuration
Switch#sh mac-address-table	To display the MAC table
Switch#sh spanning-tree	To display the STP status
Switch#sh vlan	To display VLAN status
Switch#sh flash	To display information about FLASH
Switch(config)#hostname <hostname>	To set the hostname
Switch(config)#enable password <password>	To set enable password
Switch(config)#enable secret <password>	To set enable secret password
Switch(config)#line con 0 Switch(config-line)#password <password> Switch(config-line)#login	To set line console password
Switch(config)#line vty 0 15 Switch(config-line)#password <password>	To set line VTY password
Switch(config)#username <username> password <password>	To create user and set password on it

SPANNING TREE PROTOCOL (STP)

Command	Description
Switch(config)#spanning-tree vlan 1 priority 4096	To influence root bridge election
Switch#show spanning-tree	To verify STP status
Switch(config)#int <interface name> Switch(config-if)#spanning-tree portfast disable	To configure/disable PortFast

VERIFYING & CONFIGURING VLANS (VLAN DATABASE METHOD)

Command	Description
Switch#sh vlan	To display VLAN database status
Switch#vlan database	To enter VLAN database mode
Switch(vlan)#vlan <vlan id> name <vlan name> Example: Switch(vlan)#vlan 2 name Sales	To create a VLAN
Switch(vlan)#no vlan <vlan id> Example: Switch(vlan)#no vlan 2	To delete a VLAN
Switch(vlan)#apply / exit	To save changes into VLAN
Switch(vlan)#abort	To abort changes into VLAN

ASSIGNING PORTS TO VLANS

Command	Description
Switch(config)#int <interface name> Switch(config-if)#switchport mode access Switch(config-if)#switchport access vlan <vlan id> Example: int fa0/1 -> switchport mode access -> switchport access vlan 2	Assign a single switch port into a VLAN
Switch(config)#int range <interface range> Switch(config-if-range)#switchport access vlan <vlan id> Example: int range fa0/5-8 -> switchport access vlan 3	Assign multiple switch ports into a VLAN

CONFIGURING A TRUNK PORT

Command	Description
Switch(config)#int <interface name> Switch(config-if)#switchport mode trunk Example: int fa0/1 -> switchport mode trunk	Trunk a single switch port
Switch(config)#int range <interface range> Switch(config-if-range)#switchport mode trunk Example: int range fa0/5-8 -> switchport mode trunk	Trunk multiple switch ports at a time
Switch(config)#int <interface name> Switch(config-if)#switchport mode dynamic auto	Allow port to negotiate trunk link with neighbor

CONFIGURING & VERIFYING VTP

Command	Description
Switch(config)#vtp mode <vtp mode> Example: Switch(config)#vtp mode server	To set VTP mode (server/client/transparent)
Switch(config)#vtp domain <domain name> Example: Switch(config)#vtp domain Cisco	To set VTP domain
Switch(config)#vtp password <password> Example: Switch(config)#vtp password 123	To set VTP password on domain
Switch#sh vtp status Switch#sh vtp counters	To monitor VTP operation and status

VTP PRUNING

Command	Description
Switch(config)#set vtp pruning enable	To enable VTP pruning
Switch(config)#set vtp pruneeligible 250-255	To set pruning for VLANs 500-1000
Switch#sh vtp domain	To display configuration changes

INTER-VLAN ROUTING

Command	Description
Router(config)#int fa0/0.1 Router(config-subif)#encapsulation dot1Q <vlan id> Router(config)#int fa0/0.1 Router(config-subif)#ip add <ip address> <subnet mask> Router(config-subif)#no shut	Configure router subinterface for trunk encapsulation
Router#config t Router(config)#int fa0/0 Router(config-if)#no shut Router(config)#int fa0/0.1 Router(config-subif)#encapsulation dot1Q 2 Router(config-subif)#ip add 192.168.1.254 255.255.255.0 Router(config-subif)#no shut Router(config)#int fa0/0.2 Router(config-subif)#encapsulation dot1Q 3 Router(config-subif)#ip add 192.168.2.254 255.255.255.0 Router(config-subif)#no shut	Full router-on-a-stick example (VLAN 2 & 3)

ETHERCHANNEL CONFIGURATION

Command	Description
Switch(config)#interface port-channel 1 Switch(config-if)#exit Switch(config)#interface range fa0/1-4 Switch(config-if)#switchport mode trunk Switch(config-if)#channel-group 1 mode on Switch(config-if)#switchport nonegotiate Switch(config-if)#do write	Create and configure an EtherChannel

WAN PROTOCOLS - HDLC

Command	Description
Router(config-if)#encapsulation hdlc	Set HDLC encapsulation on a serial interface (Cisco default)

WAN PROTOCOLS - PPP WITH PAP AUTHENTICATION

Two-router example (bhupal & indore).

Command	Description
Bhupal: Router(config)#hostname bhupal bhupal(config)#username indore password ccna bhupal(config)#interface s2/0 bhupal(config-if)#encapsulation ppp bhupal(config-if)#ppp authentication pap bhupal(config-if)#ppp pap sent-username bhupal password ccna	Bhupal router PPP/PAP configuration
Indore: Router(config)#hostname indore indore(config)#username bhupal password ccna indore(config)#interface s2/0 indore(config-if)#encapsulation ppp indore(config-if)#ppp authentication pap indore(config-if)#ppp pap sent-username indore password ccna	Indore router PPP/PAP configuration

WAN PROTOCOLS - PPP WITH CHAP AUTHENTICATION

Two-router example (bhupal & indore).

Command	Description
Bhupal: Router(config)#hostname bhupal bhupal(config)#username indore password ccna bhupal(config)#interface s2/0 bhupal(config-if)#encapsulation ppp bhupal(config-if)#ppp authentication chap	Bhupal router PPP/CHAP configuration
Indore: Router(config)#hostname indore indore(config)#username bhupal password ccna indore(config)#interface s2/0 indore(config-if)#encapsulation ppp indore(config-if)#ppp authentication chap	Indore router PPP/CHAP configuration